

# IMPORTANT MATH FORMULAS

## Quick Reference Guide

### ARITHMETIC

Commutative Law:  $a + b = b + a$   
 $a \cdot b = b \cdot a$   
 Associative Law:  $(a + b) + c = a + (b + c)$   
 $(a \cdot b) \cdot c = a \cdot (b \cdot c)$   
 Distributive Law:  $a(b + c) = ab + ac$   
 Identity:  $a + 0 = a$   $a \cdot 1 = a$   
 Inverse:  $a + (-a) = 0$   $a \cdot \frac{1}{a} = 1$  ( $a \neq 0$ )  
 Order of Operations: PEMDAS  
 Percent:  $p\% = \frac{p}{100}$   
 Ratio:  $a : b = \frac{a}{b}$   
 Proportion:  $\frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$

### ALGEBRA

Laws of Exponents:  
 $a^m \cdot a^n = a^{m+n}$   
 $\frac{a^m}{a^n} = a^{m-n}$  ( $a \neq 0$ )  
 $(a^m)^n = a^{mn}$   
 $(ab)^n = a^n b^n$   
 $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$  ( $b \neq 0$ )  
 $a^0 = 1$  ( $a \neq 0$ )  
 $a^{-n} = \frac{1}{a^n}$  ( $a \neq 0$ )  
 $a^{\frac{1}{n}} = \sqrt[n]{a}$

### QUADRATIC FORMULA

For  $ax^2 + bx + c = 0$ ,  

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
  
 Discriminant:  $D = b^2 - 4ac$   
 $D > 0 \rightarrow$  two real roots  
 $D = 0 \rightarrow$  one real root  
 $D < 0 \rightarrow$  no real roots

### FACTORING

$a^2 - b^2 = (a - b)(a + b)$   
 $a^2 + 2ab + b^2 = (a + b)^2$   
 $a^2 - 2ab + b^2 = (a - b)^2$   
 $ax^2 + bx + c = a(x - r_1)(x - r_2)$

### LINEAR EQUATION

Slope-intercept form:  $y = mx + b$   
 Point-slope form:  $y - y_1 = m(x - x_1)$   
 Slope:  $m = \frac{y_2 - y_1}{x_2 - x_1}$   
 Standard form:  $Ax + By = C$

### SYSTEM OF LINEAR EQUATIONS

$\left. \begin{array}{l} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{array} \right\}$  • Substitution  
 • Elimination  
 • Graphing

### POLYNOMIALS

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

Remainder Theorem:

If  $f(x)$  is divided by  $(x - c)$ ,  
 the remainder is  $f(c)$ .

Factor Theorem:

If  $(x - c)$  is a factor of  $f(x)$ ,  
 then  $f(c) = 0$ .

### SEQUENCES AND SERIES

Arithmetic Sequence:

$$a_n = a_1 + (n - 1)d$$

$$\text{Sum: } S_n = \frac{n}{2} [2a_1 + (n - 1)d]$$

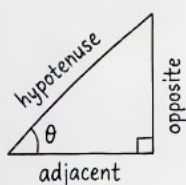
Geometric Sequence:

$$a_n = a_1 r^{n-1}$$

$$\text{Sum } (r \neq 1): S_n = \frac{a_1(1 - r^n)}{1 - r}$$

$$\text{Sum to infinity } (|r| < 1): S = \frac{a_1}{1 - r}$$

### TRIGONOMETRY



$$\begin{aligned} \sin \theta &= \frac{\text{opposite}}{\text{hypotenuse}} \\ \cos \theta &= \frac{\text{adjacent}}{\text{hypotenuse}} \\ \tan \theta &= \frac{\text{opposite}}{\text{adjacent}} \\ \csc \theta &= \frac{1}{\sin \theta} \\ \sec \theta &= \frac{1}{\cos \theta} \\ \cot \theta &= \frac{1}{\tan \theta} \end{aligned}$$

Special Angles:

| $\theta$   | $\sin \theta$        | $\cos \theta$        | $\tan \theta$        |
|------------|----------------------|----------------------|----------------------|
| $0^\circ$  | 0                    | 1                    | 0                    |
| $30^\circ$ | $\frac{1}{2}$        | $\frac{\sqrt{3}}{2}$ | $\frac{1}{\sqrt{3}}$ |
| $45^\circ$ | $\frac{\sqrt{2}}{2}$ | $\frac{\sqrt{2}}{2}$ | 1                    |
| $60^\circ$ | $\frac{\sqrt{3}}{2}$ | $\frac{1}{2}$        | $\sqrt{3}$           |
| $90^\circ$ | 1                    | 0                    | undefined            |

Pythagorean Identity:

$$\sin^2 \theta + \cos^2 \theta = 1$$

Other Identities:

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

### GEOMETRY

$$\text{Triangle: } A = \frac{1}{2}bh$$

$$\text{Rectangle: } A = lw \quad P = 2(l + w)$$

$$\text{Parallelogram: } A = bh$$

$$\text{Trapezoid: } A = \frac{1}{2}(b_1 + b_2)h$$

$$\text{Circle: } A = \pi r^2 \quad C = 2\pi r$$

$$\text{Sector: } A = \frac{\theta}{360^\circ} \pi r^2$$

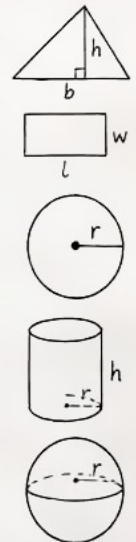
Surface Area (Cylinder):

$$S = 2\pi rh + 2\pi r^2$$

$$\text{Volume (Cylinder): } V = \pi r^2 h$$

$$\text{Surface Area (Sphere): } S = 4\pi r^2$$

$$\text{Volume (Sphere): } V = \frac{4}{3}\pi r^3$$



### COORDINATE GEOMETRY

Distance Formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Midpoint Formula:

$$M \left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$\text{Slope: } m = \frac{y_2 - y_1}{x_2 - x_1}$$

### STATISTICS

$$\text{Mean: } \bar{x} = \frac{\sum x}{n}$$

Median: (middle value)

Mode: (most frequent value)

Range:  $R = \text{largest} - \text{smallest}$

$$\text{Variance (sample): } s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$

$$\text{Standard Deviation (sample): } s = \sqrt{s^2}$$

### LOGARITHMS

$$\log_b(xy) = \log_b x + \log_b y$$

$$\log_b \left( \frac{x}{y} \right) = \log_b x - \log_b y$$

$$\log_b(x^n) = n \log_b x$$

$$\log_b x = y \Leftrightarrow b^y = x$$

$$\text{Change of base: } \log_b x = \frac{\log_k x}{\log_k b}$$

"Practice today for a brighter tomorrow!" 😊